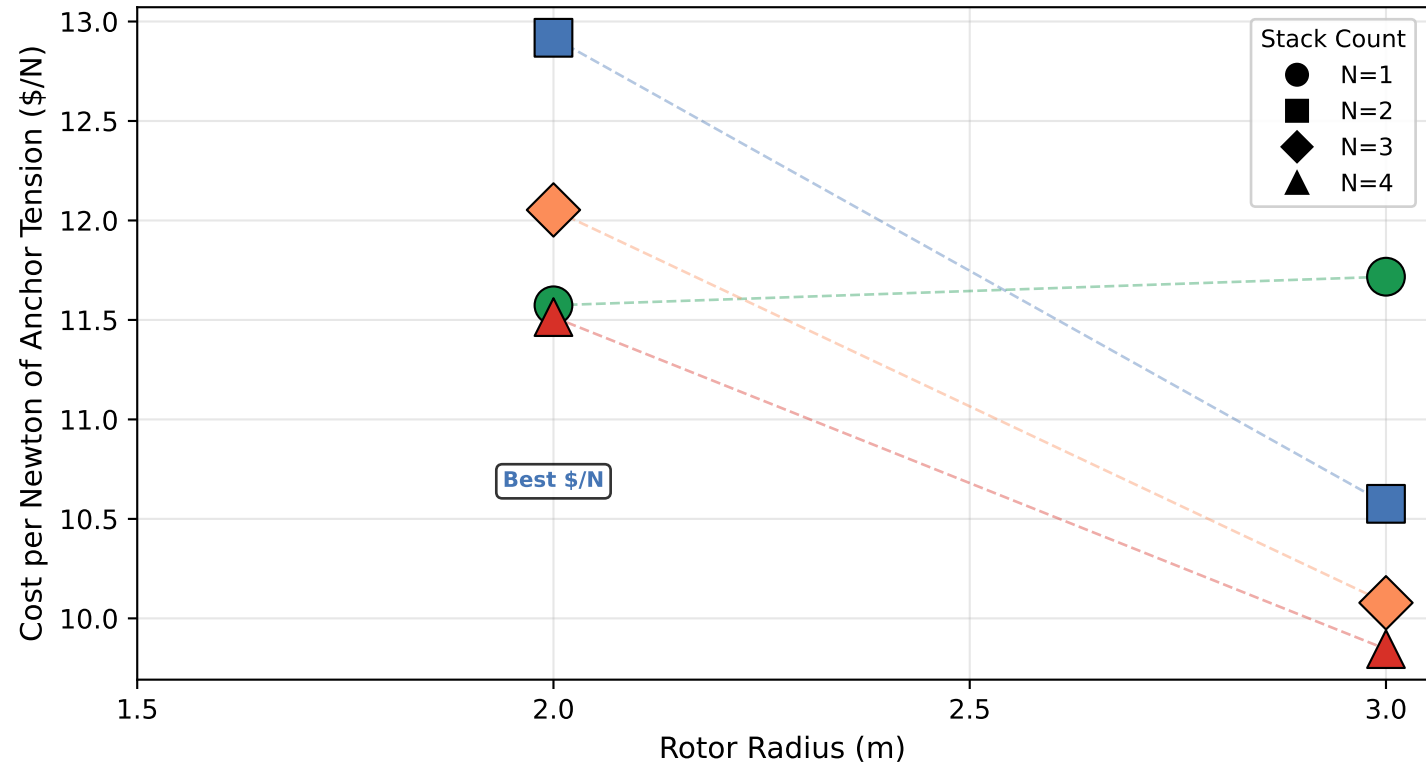


Cost per Newton: Economics of Scale

BEM v2.1 — Estimated Manufacturing Cost Model



IMPLICATIONS: Larger rotors ($R=3.0\text{m}$) have lower cost per newton despite higher unit cost because tension scales with R^2 while cost scales with R (material). $N=1$ has the best $\$/N$ ratio but delivers too little absolute tension. $N=3-4$ provides the best balance of cost efficiency and absolute performance. The cost model is simplified — real costs depend on manufacturing volume, material choices, and labour. Data: BEM v2.1, graded profile.